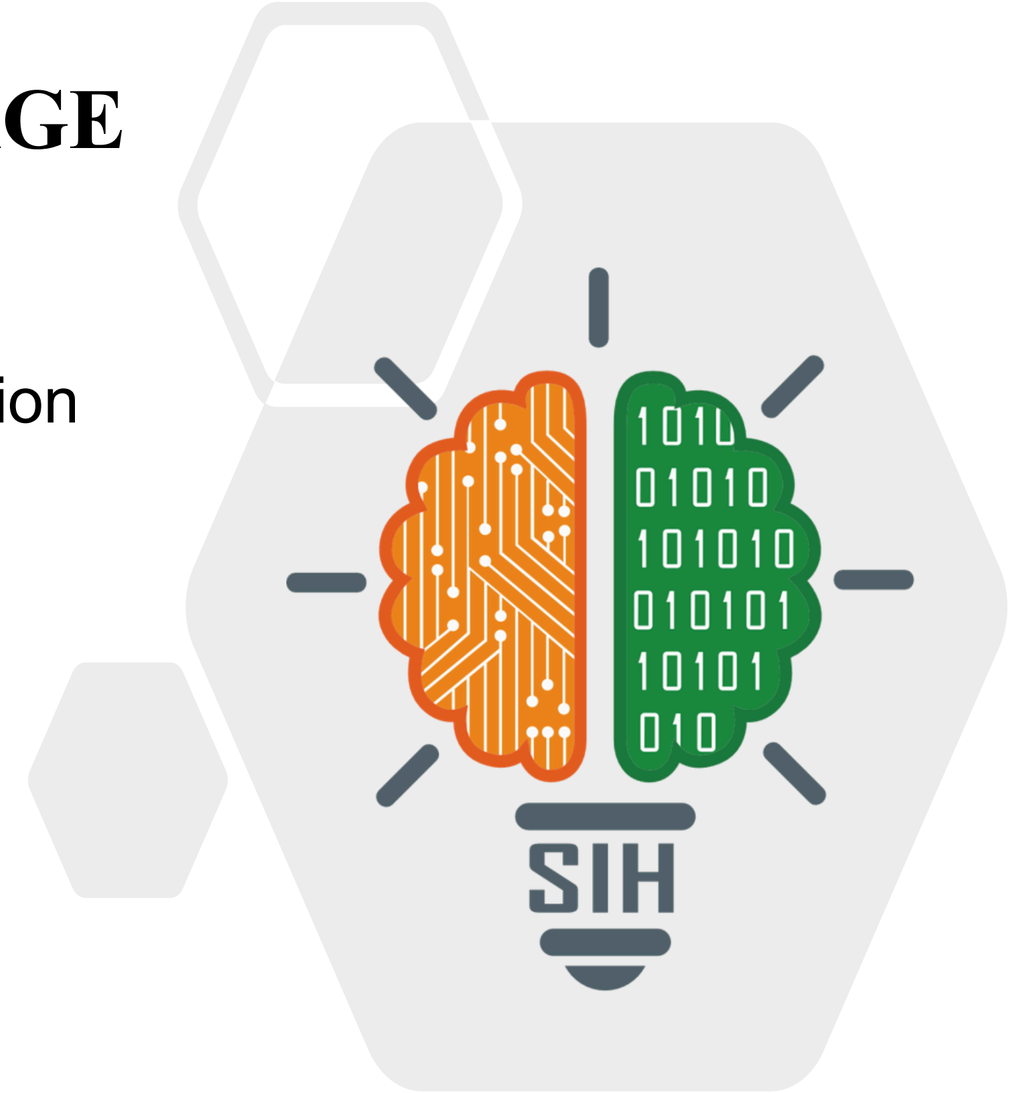
TITLE PAGE

- **Problem Statement ID:** SIH162
- **Problem Statement Title:** AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data
- **Theme:** Miscellaneous
- **PS Category:** Software
- **Team ID:** SIH26-DU-T07
- **Team Name:** AstraX



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IGNISSENSE

AI & SATELLITE-POWERED INDUSTRIAL FIRE INTELLIGENCE PLATFORM

SMART INDIA HACKATHON 2026

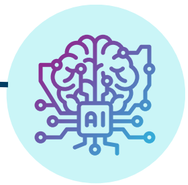
PROPOSED SOLUTION



Satellite Data Ingestion:
Automatically takes in estimated real-time NASA satellite fire data (Brightness Temperature and Fire Radiative Power)



Geospatial Context Fusion:
Maps real-time fire warnings to nearby industrial infrastructure using MongoDB spatial indexing + geospatial queries



AI False Alarm Filtering:
Apply XGBoost algorithm to distinguish chimney fires from true incidents, based on thermal oddity patterns



Incident Command Dashboard:
Visualize high-risk areas in real-time with single-click GeoJSON export for emergency response teams

KEY INNOVATIONS AND UNIQUENESS



Class ML & Explainable AI: Categorises risks and elaborates on alerts with SHAP insights



Dual-Layer Logic & Interoperability: Minimises count of false alarms with persistence baselines; exports GeoJSON/CSV for QGIS and state systems.

ACTIONABLE INTELLIGENCE



Risk Based Prioritization:
Ranks fire alerts using thermal intensity, proximity

Automated Response Alerts:
Triggers prioritized notifis with location, risk level, etc

ADDRESSING THE PROBLEM



Zero-Lag & Zero CapEx: Identifies satellite heat signatures within minutes, safeguarding industrial corridors without the use of expensive ground sensors.



Context-Aware Triage & Scale: Computes hazard proximity to critical architecture for prioritized dispatch—scalable nationwide.

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Technical Approach

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01 DATA SOURCES

Free, public, updated by others

NASA FIRMS

Area API, near-real-time

- VIIRS_SNPP_NRT, VIIRS_NOAA20_NRT, VIIRS_NOAA21_NRT (375 m)
- MODIS_NRT on Terra, Aqua (1 km)
- CSV over HTTPS with a MAP_KEY
- Last 2 days per poll (API allows 1–5)
- India rectangle 68–97.5°E, 6.5–37.5°N

OpenStreetMap

Geofabrik India extract

- 341,086 features across India
- Industry, power, oil & gas, mining
- Same tag filter as the Overpass extractor

02 INGESTION

Backend · Node.js, Express 5, TS · :5001

1 Pull satellite data

firms.client.ts · scheduler.ts

- 4 FIRMS products, fetched one after another
- Every 5 min; runs never overlap; catch-up after a restart
- 4 requests per cycle; limit is 5,000 per 10 min
- One IngestionLog row per product

2 Clean and deduplicate

ingestion.service.ts · indiaBoundary.ts

- VIIRS I-4/I-5 and MODIS 21/31 mapped to one schema
- Malformed rows dropped
- Point-in-polygon test against India's boundary
- Unique index on location, time, satellite, sensor
- 457,869 repeat rows rejected so far

03 ENRICH AND FEATURES

Backend · MongoDB 2dsphere queries

3 Spatial enrichment

enrichment.service.ts · taxonomy.ts

- Nearest OSM feature within 25 km ($nearSphere, 2dsphere); closest category wins
- 9 categories: industrial, oil & gas, mining, power, forest, agriculture, urban, water, other
- Land cover inferred from that category

4 Build feature vector

feature.extractor.ts · featureContract.ts

- 14 values per detection
- Radiometry: FRP, TI4, TI5, ΔT, confidence, day/night
- Context: facility distance and type, land cover
- History: 3 km / 72 h cluster, 1.5 km recurrence, mean FRP, z-score, days since last
- New detections classified 15 at a time

5 Completeness check

classification.service.ts

- 10 of the 14 must resolve; the 4 history features may be empty
- If not: stored as **unclassified**, no model call (229 so far)

04 CLASSIFY

Inference · Python, XGBoost 3.4 · :8000

6 Classify

service.py · feature_frame.py

- POST /predict on a stdlib HTTP server
- HTTP 400 if a required feature is missing
- Empty history → 0; days_since stays NaN; same code as train.py
- XGBoost XGB-FIRMS-v2.0.0-INDIA returns the class and all 6 probabilities
- Service down: no prediction, row tagged -OFFLINE (153 so far)

How it was trained

build_labels.py · train.py

- FIRMS archive 2022–24, independent labels
- 0.5° blocks; train 2023, test 2024
- Macro-F1 0.55 on held-out 2024 regions (distance rule 0.41)

05 SCORE AND ALERT

Backend · rules in TypeScript

7 Score and raise alerts

classification.service.ts

- Persistence and anomaly scores (0–1), rule-based
- Alert when anomaly ≥ 0.65 and the facility is ≤ 1.5 km
- Critical at anomaly ≥ 0.80
- Explanation text built from the features

Alert lifecycle

alert.model.ts · AlertsPage

- open → acknowledged → resolved, set by an officer
- Sort by latest, severity or anomaly; filter by severity
- 74 alerts so far: 34 critical, 37 high, 3 medium

06 STORE AND SERVE

Atlas · React 19 · :5173

MongoDB Atlas

geospatial store

- hotspots
- classifications
- alerts
- osmfeatures
- osmtiles
- facilities
- ingestionlogs
- geocodecaches

2dsphere on hotspots and osmfeatures; unique dedup index

8 Serve the map

app.ts · MapView.tsx

- REST /api/v1 with a {success, data, error} envelope; Helmet; 500 req / 15 min
- MapLibre GL map with layers and filters
- Unclassified shown as hollow rings
- Search coordinates, places, facilities
- Detail panel: facility, district, FRP history
- TanStack Query refresh every 15–60 s

Thermal-Event Classes — training label rule

Industrial Fire	Static source, no flare or mine
Gas Flare	Static source ≤ 1 km from a flare site
Wildfire	Vegetation fire, ≥ 60% tree/grass
Agricultural Burning	Vegetation fire, ≥ 60% cropland
Mining Thermal Activity	Static source inside a mining polygon
Other / Uncertain	Volcano, offshore, mixed cover

Feature Set — one schema, 14 features

Required 10 must be present

- frp
- brightness
- brightness_ti5
- temp_delta_ti4_ti5
- confidence
- is_night
- facility_distance_m
- facility_type_encoded
- landcover_encoded
- nearby_cluster_count_3km

Optional 4 can be absent

- historical_recurrence_1_5km
- historical_mean_frp
- frp_z_score
- days_since_last_detection

A Vitest test parses classes.py and fails if the TypeScript and Python feature lists drift apart.

Current System Metrics — live database, 26 Sep 2026

12,966	satellite detections
12,569	classified by the model
341,086	OSM features indexed
73	open alerts
457,869	repeat rows rejected

Known Limitations — from model metadata

- **Labels are rule-based.** From FIRMS type, flare sites, mine maps and WorldCover, not field visits.
- **No facility within 25 km, no class.** Remote wildfires are under-represented in training and on the map.
- **Gas flare F1 is 0.24.** India has only about 156 known flare sites to learn from.
- **Every sensor run is logged** with status and counts (1,662 runs so far).

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Feasibility and Viability

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Analysis of the feasibility of the idea

measured on the running prototype

4

Satellites

VIIRS on S-NPP, NOAA-20 and NOAA-21, plus MODIS, give several passes over India every day.

4 / 5,000

FIRMS requests

4 requests per 5-minute cycle against 5,000 allowed per 10 minutes. An average run takes 42 s.

341,086

Facilities mapped

Industrial, power, oil & gas and mining sites from OpenStreetMap, at no cost.

0.55

Macro-F1

On 2024 regions never seen in training, against 0.41 for a distance rule. Trains on a CPU.

71

Alerts raised

Each near a mapped plant, with its reasons, facility distance and class probabilities.

0

Installs needed

The dashboard runs in a browser. 77 automated tests pass across the backend.



Potential challenges and risks

where each stage could break

- **Clouds and smoke hide fires, and 375 m pixels can miss small burns.**
- **FIRMS serves only the last 5 days, so a longer outage loses data. It happened on 25–28 Aug.**
- **No mapped facility within 25 km means no class. 219 detections so far.**
- **Labels come from public datasets by rule, not field checks. Gas flares are weakest (F1 0.24).**
- **A wrong alert could lead to action against the wrong plant or farmer.**
- **Officers already have routines and may not take up a new tool.**

Strategies for overcoming these challenges

the fix for that stage

- Merge all four sensors and judge persistence over 72 h rather than single passes. Confirm important sites with Sentinel-2 imagery.
- Run on an always-on server. On restart, catch-up refills the missed window by itself. Backfill older gaps from FIRMS archive files.
- Add CPCB and state industry registers to the facility layer, refresh OSM monthly, and re-run gated detections after each import.
- Pilot with a State PCB so inspection results become labels, retrain each season, and use the EOG flare list as a direct lookup.
- An officer acknowledges or resolves every alert. Nothing goes straight to enforcement, and unclear cases stay unclassified.
- Start with one district. The alert list works like an inspection queue, and staff get hands-on training during the pilot.

Viability

why it keeps running

No paid data or hardware: FIRMS, OpenStreetMap and WorldCover are free, and nothing is installed in the field.
Small to run: three open-source services and one MongoDB database.
Already India-wide: a new state needs its facility registers, not a new model.

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Impacts and Benefits

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POTENTIAL IMPACT ON THE TARGET AUDIENCE

Agriculture and district offices

Amritsar, Punjab

1,802 detections classed as crop burning

85 near Amritsar as the paddy-residue season begins. Each is tagged with its district, so action goes to the right office.

Oil and gas regulators

Hazira, Gujarat

105 gas flares

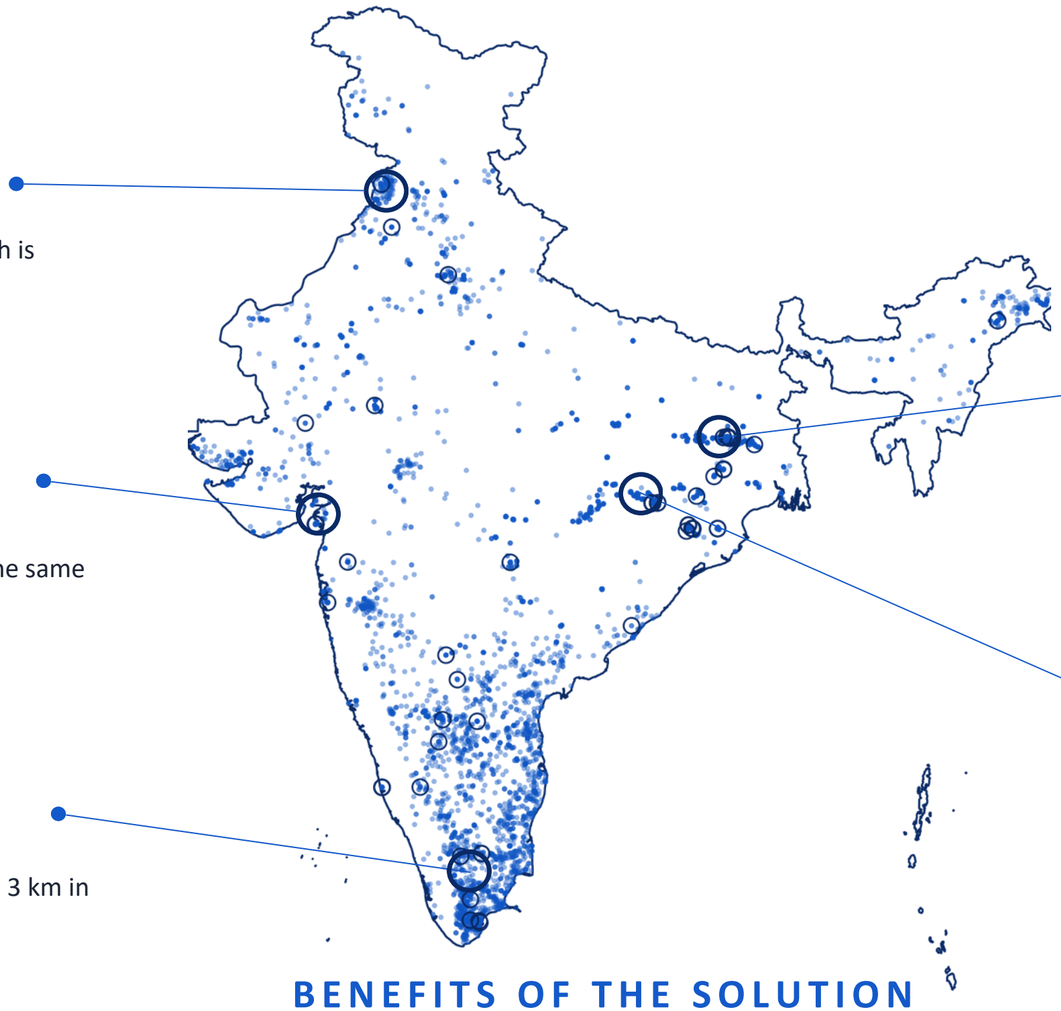
32 around Hazira's gas processing plants. Recurrence at the same 1.5 km spot separates routine flaring from a new event.

Forest departments and SDMAs

Western Ghats, Kerala and Tamil Nadu

1,749 wildfire detections

137 in the southern Ghats. The count of detections within 3 km in 72 h shows which fires are spreading.



- **12,585** FIRMS detections, 20 Aug – 23 Sep 2026, from the running system; 97.1% classified automatically
- **71** alerts: high anomaly within 1.5 km of a mapped plant

Mining regulators

Jharia coalfield, Jharkhand

1,611 mining heat detections

691 over Jharia, where underground coal fires have burned for decades. FRP history shows which sites are getting hotter.

Pollution control boards

Raigarh–Jharsuguda, Chhattisgarh and Odisha

1,180 industrial fires

191 in this steel and aluminium belt. Near a mapped plant they become alerts with facility, distance and reasons.

BENEFITS OF THE SOLUTION

Social

Earlier flags give district health officials time to issue smoke advisories, even where there is no ground air-quality monitor. Alerts name the district and state in plain words.

Economic

Built only on free data: NASA FIRMS, OpenStreetMap, ESA WorldCover. No sensors to buy or install. Field teams start from a ranked alert list, not a raw map.

Environmental

Action goes to the actual source: crop, forest, flare, mine or plant. Repeat detections at one spot expose chronic emitters, not just one-off fires.

Governance

Every alert shows its reasons and class probabilities. 219 detections with no mapped facility were left unclassified rather than guessed. MIT-licensed and open to audit.

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SMART INDIA HACKATHON 2026

Research and References

Research and Related Work



Thermal anomaly detection
Uses satellite-based thermal data to detect active fire events and thermal anomalies



Geospatial context
Provides geospatial context such as locations of industrial facilities, roads and infrastructure



Land Cover & Historical Data
Helps identifying land use patterns and analyse past thermal behaviour



XGBoost & Explainable AI
Effective for multi-class classification and provides feature importance for transparency

References



NASA FIRMS
Fire information for resource management system
https://firms.modaps.eosdis.nasa.gov/



OpenStreetMap
Open data for geospatial information
https://www.openstreetmap.org/#map=2/-41.2/-6.6



MongoDB Atlas
Cloud database for geospatial data
https://www.mongodb.com/products/platform/atlas-database



Copernicus EMS
Disaster mapping & emergency response
copernicus.eu/en/copernicus-services

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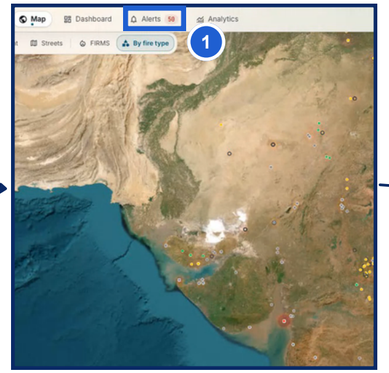
User Flow

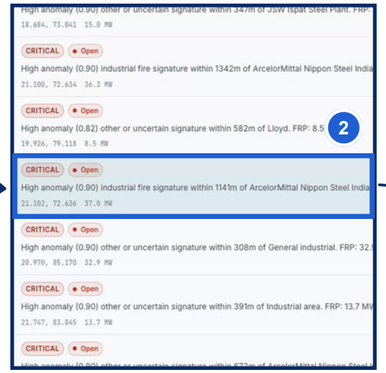
SMART INDIA HACKATHON 2026

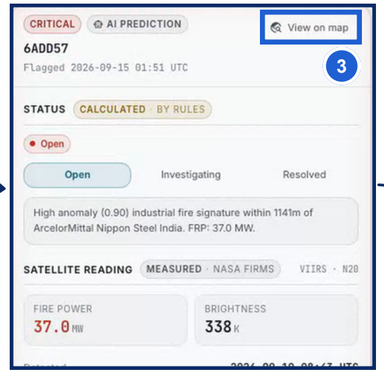
Github Repo Link

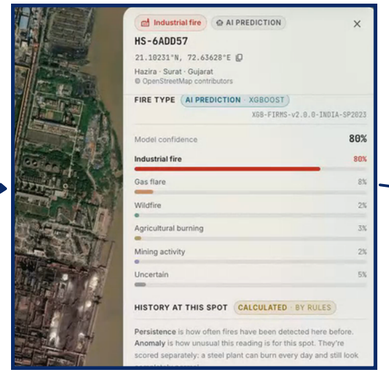
USER	GOAL	STARTS AT	SUCCESS
Fire monitoring officer	Check a flagged fire, record a decision	Open-alert count on the map	Alert marked Investigating or Resolved

START
Opens the app









05
Fire?

YES
Investigating

NO
Resolved

NOT SURE
Leave open

	01 Live map	02 Alert list	03 Alert detail	04 Detection on map	05 Decide
DOES	Sees the open-alert count on the Alerts tab	Filters to critical alerts and picks the ArcelorMittal one	Reads why it was flagged, then taps View on map	Checks fire type, the nearest site and past fires here	Sets the status back on the alert. The open count drops.
ASKS	*"Anything new since I last looked?"*	*"Which one needs me first?"*	*"Why did this get flagged?"*	*"Is this really a factory fire?"*	*"Send a team, or close it?"*

Not sure: the alert stays open, the next satellite pass adds history to the spot, and the officer checks again

start or end · screen · decision · 1 where the officer clicks

BEHIND THE SCREENS New NASA FIRMS detections every 5 minutes. XGBoost names the fire type, and a fixed rule raises the alert: anomaly ≥ 0.65 with a mapped site within 1.5 km.